

Institute of Science and Technology
BernHardt College
Bafal, Kathmandu

Subject: Compiler Design and Construction (CSC365)

FM:60

Bachelor Six Semester

PM: 24

Candidates are required to give their answer in their own words as far as practicable.

Time: 3hr

Section A

Attempt any two questions. (2*10=20)

1. Describe the phases of compiler with block diagram and an example.
2. Differentiate between top-down and bottom-up parsing. Construct SLR parse table for following given grammar:

$S \rightarrow aETe$

$E \rightarrow Ebc$

$E \rightarrow b$

$T \rightarrow d$

3. What are the roles of lexical syntax and semantic analyzer. Define type checking. Different between type conversion and coercion.

Section B

Attempt any eight questions. (8*5=40)

1. What is semantic analyzer? What errors are generated during semantic analyzing? (2+3)
2. What is input buffering how it works? (5)
3. Define DFA. Convert the Regular Expression $(a+b)^*aab$ to DFA directly. (1+4)
4. Define left recursive grammar. Remove left recursion from the following grammar. (1+4)
 $S \rightarrow SB \mid Ca$
 $B \rightarrow Bb \mid c$
 $C \rightarrow aB \mid a$
5. What are the disadvantages of shift reduce parsing? Perform shift reducing parsing of string: (1+4)

$(x-x) - (x/x)$

Grammar: $E \rightarrow E-E \mid E/E \mid (E) \mid x$

6. Find first and follow of all the non-terminals in the given grammar. (5)

$A \rightarrow TE$

$E \rightarrow +TE \mid \epsilon$

$T \rightarrow XY$

$Y \rightarrow *XY \mid \epsilon$

$X \rightarrow (A) \mid a$

7. Construct SLR parse table for: (5)

$S \rightarrow XX$

$X \rightarrow pX \mid q$

8. Show that the following grammar is not in a LL (1): (5)

$S \rightarrow cAd$

$A \rightarrow Ab \mid a$

9. Write short notes on: (2.5*2 = 5)

a. Handle

b. Symbol Table